

Liutang Xue

Curriculum Vitae

Professor of Mathematics

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Academic Positions

08/2022–Present	Professor , Beijing Normal University, Beijing, China.
09/2016–07/2022	Associate Professor , Beijing Normal University, Beijing, China.
10/2013–09/2016	Assistant Professor , Beijing Normal University, Beijing, China.
09/2012–09/2013	Postdoctoral Researcher , Laboratoire LAMA, Université Paris-Est, France. Advisors: Prof. Régis Monneau and Prof. Marco Cannone.
02/2017–05/2017	Visiting Researcher , University of California, Irvine, USA.

Education

09/2009–07/2012	Ph.D. in Applied Mathematics , Graduate School of China Academy of Engineering Physics, Beijing, China. Advisor: Prof. Changxing Miao.
09/2006–07/2009	M.S. in Applied Mathematics , Graduate School of China Academy of Engineering Physics, Beijing, China. Advisor: Prof. Changxing Miao.
09/2002–07/2006	B.S. in Mathematics and Applied Mathematics , Zhengzhou University, Henan, China.

Research Interests

Analysis of nonlinear partial differential equations; harmonic analysis; mathematical fluid mechanics, including the Euler, Navier–Stokes, Boussinesq, SQG, and Euler-alignment equations.

Research Grants

National Natural Science Foundation of China (NSFC); Principal Investigator for all grants.

01/2027–12/2030	General Program , No. 12671270 .
01/2023–12/2026	General Program , No. 12271045 .
01/2018–12/2021	General Program , No. 11771043 .
01/2015–12/2017	Young Scientists Fund , No. 11401027 .

Publications and Preprints

- [1] Changhui Tan and **Liutang Xue**. Global regularity for the unidirectional Euler-alignment system with supercritical dissipation. Preprint. [arXiv:2609.25748](#).
- [2] Changhui Tan, **Liutang Xue**, and Zhilong Xue. A revisit of patch solutions for the 2D Loglog-Euler type equation. To appear in *Revista Matemática Iberoamericana*. [arXiv:2510.18759](#).
- [3] Qianyun Miao, Changhui Tan, **Liutang Xue**, and Zhilong Xue. Local regularity and finite-time singularity for a class of generalized SQG patches on the half-plane. Preprint. [arXiv:2410.19273](#).
- [4] Taoufik Hmidi, **Liutang Xue**, and Zhilong Xue. Doubly connected V-states in geophysical models: a general framework. *Calc. Var. Partial Differential Equations*, 65 (2026), Art. No. 137.
- [5] Taoufik Hmidi, **Liutang Xue**, and Zhilong Xue. Unified theory on V-states structures for active scalar equations. *Adv. Math.*, 486 (2026), Art. No. 110750.
- [6] Omar Lazar, **Liutang Xue**, and Jiakun Yang. Global regularity and infinite Prandtl number limit of temperature patches for the 2D Boussinesq system. *Mathematische Annalen*, 392 (2025), 1375–1446.
- [7] Yatao Li, Qianyun Miao, Changhui Tan, and **Liutang Xue**. Global well-posedness and refined regularity criterion for the uni-directional Euler-alignment system. *International Mathematics Research Notices*, 2024 (2024), no. 23, 14393–14422.
- [8] Xiang Bai, Changhui Tan, and **Liutang Xue**. Global well-posedness and asymptotic behavior for the Euler-alignment system with pressure. *J. Differential Equations*, 407 (2024), 269–310.
- [9] Xiang Bai, Qianyun Miao, Changhui Tan, and **Liutang Xue**. Global well-posedness and asymptotic behavior in critical spaces for the compressible Euler system with velocity alignment. *Nonlinearity*, 37 (2024), no. 2, Paper No. 025007.
- [10] Omar Lazar, Yatao Li, and **Liutang Xue**. On the regularity of temperature fronts for the 3D viscous Boussinesq system. *Indiana Univ. Math. J.*, 73 (2024), no. 3, 1005–1060.
- [11] Taoufik Hmidi, **Liutang Xue**, and Zhilong Xue. Emergence of time periodic solutions for the generalized surface quasi-geostrophic equation in the disc. *Journal of Functional Analysis*, 285 (2023), no. 10, Paper No. 110142.
- [12] Yatao Li, Qianyun Miao, and **Liutang Xue**. Global well-posedness for 2D fractional inhomogeneous Navier-Stokes equations with rough density. *Nonlinearity* 36 (2023), 3866–3908.
- [13] Dongho Chae, Qianyun Miao, and **Liutang Xue**. Global regularity of non-diffusive temperature fronts for the 2D viscous Boussinesq system. *SIAM J. Math. Anal.* 54 (2022), no. 4, 4043–4103. *Erratum*: 56 (2024), no. 6, 8125–8130.
- [14] Piotr B. Mucha and **Liutang Xue**. Boussinesq system with measure forcing. *Mathematische Annalen*, 382 (2022), 1901–1934.
- [15] Qianyun Miao, Changhui Tan, and **Liutang Xue**. Global regularity for a 1D Euler-alignment system with misalignment. *Math. Mod. Meth. Appl. Sci.*, 31 (2021), no. 3, 473–524.
- [16] Qianyun Miao and **Liutang Xue**. Regularity and singularity results for the dissipative Whitham equation and related surface wave equations. *Commun. Math. Sci.*, 17 (2019), no. 8, 2141–2190.
- [17] Omar Lazar and **Liutang Xue**. Regularity results for a class of generalized surface quasi-geostrophic equations. *Journal de Mathématiques Pures et Appliquées*, 130 (2019), 200–250.
- [18] Piotr B. Mucha, **Liutang Xue**, and Xiaoxin Zheng. Between homogeneous and inhomogeneous Navier-Stokes systems: the issue of stability. *J. Differential Equations*, 267 (2019), Issue 1, 307–363.
- [19] Changxing Miao and **Liutang Xue**. On the regularity issues of a class of drift-diffusion equations with nonlocal diffusion. *SIAM J. Math. Anal.*, 51 (2019), no. 4, 2927–2970.
- [20] Qianyun Miao and **Liutang Xue**. On the Hölder regularity of the weak solution to a drift-diffusion equation with pressure. *Calc. Var. Partial Differential Equations*, 57 (2018), no. 6, Art. 153, 33 pp.
- [21] **Liutang Xue** and Zhuan Ye. On the differentiability issue of the drift-diffusion equation with nonlocal Lévy-type diffusion. *Pacific J. Mathematics* 293 (2018), no. 2, 471–510.

- [22] Zhuan Ye, Xiaojing Xu, and **Liutang Xue**. On the global regularity of the 2D Boussinesq equations with fractional dissipation. *Math. Nachr.* 290 (2017), nos. 8–9, 1420–1439.
- [23] Jiahong Wu, Xiaojing Xu, **Liutang Xue**, and Zhuan Ye. Regularity results for the 2D Boussinesq equations with critical or supercritical dissipation. *Commun. Math. Sci.* 14 (2016), no. 7, 1963–1997.
- [24] **Liutang Xue**. On the locally self-similar solution of the surface quasi-geostrophic equation with decaying or non-decaying profiles. *J. Differential Equations* 261 (2016), no. 10, 5590–5608.
- [25] **Liutang Xue**. On the locally self-similar singular solutions for the incompressible Euler equations. *Dyn. Partial Differ. Equ.* 12 (2015), no. 4, 321–342.
- [26] **Liutang Xue**. On the discretely self-similar solutions for the 3D Navier-Stokes equations. *Nonlinearity* 28 (2015), no. 10, 3695–3708.
- [27] Marco Cannone and **Liutang Xue**. Remarks on self-similar solutions for the surface quasi-geostrophic equation and its generalization. *Proc. Amer. Math. Soc.* 143 (2015), no. 6, 2613–2622.
- [28] Dong Li, Changxing Miao, and **Liutang Xue**. On the well-posedness of a 2D nonlinear and nonlocal system arising from the dislocation dynamics. *Commun. Contemp. Math.* 16 (2014), no. 2, 1350021, 36 pp.
- [29] Xiaojing Xu and **Liutang Xue**. Yudovich type solution for the 2D inviscid Boussinesq system with critical and supercritical dissipation. *J. Differential Equations* 256 (2014), no. 9, 3179–3207.
- [30] Marco Cannone, Changxing Miao, and **Liutang Xue**. Global regularity for the supercritical dissipative quasi-geostrophic equation with large dispersive forcing. *Proc. Lond. Math. Soc.* (3) 106 (2013), no. 3, 650–674.
- [31] **Liutang Xue** and Xiaoxin Zheng. Note on the well-posedness of a slightly supercritical surface quasi-geostrophic equation. *J. Differential Equations* 253 (2012), no. 2, 795–813.
- [32] Gang Wu and **Liutang Xue**. Global well-posedness for the 2D inviscid Bénard system with fractional diffusivity and Yudovich’s type data. *J. Differential Equations* 253 (2012), no. 1, 100–125.
- [33] Changxing Miao and **Liutang Xue**. Global well-posedness for a modified critical dissipative quasi-geostrophic equation. *J. Differential Equations* 252 (2012), no. 1, 792–818.
- [34] Changxing Miao and **Liutang Xue**. On the regularity of a class of generalized quasi-geostrophic equations. *J. Differential Equations* 251 (2011), no. 10, 2789–2821.
- [35] Changxing Miao and **Liutang Xue**. On the global well-posedness of a class of Boussinesq-Navier-Stokes systems. *NoDEA Nonlinear Differential Equations Appl.* 18 (2011), no. 6, 707–735.
- [36] **Liutang Xue**. Wellposedness and zero microrotation viscosity limit of the 2D micropolar fluid equations. *Math. Methods Appl. Sci.* 34 (2011), no. 14, 1760–1777.
- [37] **Liutang Xue**. On the well-posedness of incompressible flow in porous media with supercritical diffusion. *Appl. Anal.* 88 (2009), no. 4, 547–561.

Selected Invited Talks and Conference Presentations

- **6–10 July 2026.** 15th AIMS Conference on Dynamical Systems, Differential Equations and Applications. Athens, Greece. Special Session 22: Models of Emergence and Collective Dynamics.
- **17–19 April 2026.** Workshop on Fluid Mechanics Equations. Beijing Information Science and Technology University, Beijing, China. Talk on 19 April.
- **10–13 April 2026.** Workshop on Singularity and Stability of Incompressible Fluid Equations. Hangzhou Normal University, Hangzhou, China. Talk on 11 April.
- **6–8 March 2026.** Workshop on Frontiers in the Mathematical Theory of Incompressible Fluid Equations. Zhejiang Normal University, Jinhua, China. Talk on 7 March.
- **21–26 September 2025.** MathFlows'25. Porquerolles, France. Invited talk on 23 September. *Regularity and finite-time singularity for a class of generalized SQG patches on the half-plane.*
- **16–20 December 2024.** 14th AIMS Conference. NYU Abu Dhabi, Abu Dhabi, United Arab Emirates.

- **8–12 July 2024.** Alhambra PDE Days, satellite conference of the 9th European Congress of Mathematics. University of Granada, Granada, Spain.
- **3–7 June 2024.** KinMAT Workshop. Warsaw, Poland.
- **30–31 December 2023.** Mathematical Theory of Fluid Equations and Related Models. Beihang University, Beijing, China.
- **15–16 July 2023.** Workshop on PDEs in Fluid Mechanics. Zhengzhou University of Light Industry, China.
- **10–11 December 2022.** Partial Differential Equations Theory and Applications. Central China Normal University, China (online).
- **30 September 2022.** Analysis & PDE Seminar. Academy of Mathematics and Systems Science, Chinese Academy of Sciences, Beijing, China.
- **3 March 2022.** Fluid Conversation Seminar. University of Seville, Spain (online).
- **30 November–3 December 2021.** International Workshop on Multiphase Flows: Analysis, Modelling and Numerics. Waseda University, Japan (online).
- **16–17 October 2021.** Recent Developments in Nonlinear PDEs in Fluid Mechanics and Related Fields. Zhejiang Normal University, Jinhua, China.
- **28–30 November 2020.** 7th PDEs Youth Scholars Forum. South China Normal University, Guangzhou, China.
- **8–13 March 2020.** Mathflows 2020. Będlewo, Poland. *Invitation; conference cancelled.*
- **6 July 2019.** Workshop on Mathematical Theory of Fluid Mechanics. Beijing Normal University, Beijing, China.
- **3 September 2018.** Mathflows 2018. Porquerolles, France.
- **11 April 2017.** PDE Seminar. University of California, Irvine, USA.
- **10 May 2016.** PDE Seminar. Chung-Ang University, South Korea.
- **24–25 January 2015.** 5th BNU-PDE Workshop. Beijing Normal University, Beijing, China.
- **10 January 2013.** PDE Seminar. Rennes 1 University, France.
- **25 June–13 July 2012.** Dynamics of Nonlinear Dispersive and Fluid Mechanics Equations. Beijing International Center for Mathematical Research (BICMR), Beijing, China.
- **17–23 May 2010.** Beijing Conference on Harmonic Analysis and Partial Differential Equations. Institute of Applied Physics and Computational Mathematics (IAPCM), Beijing, China.